BILL OF MATERIALS

Modular Mini Conveyor Belt for Arduino Automation

Rev. 2026-10-20 16 items

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BOM	Product/Project: Modular Mini Conveyor Belt for Arduino Automation Revision date: 2026-10-20	https://www.printables.com/@TomKnox_1573/models

MOTION

1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	TT DC gear motor 6V metal gearbox 1:90 with shaft	**1**	AliExpress Link	Amazon Link	*Upstream lists a TT motor 3-6 V with metal gearbox recommended. Pick the dual-shaft metal-gear version; the plastic-gear kit motor*

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ELECTRONICS 1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	L9110S dual H-bridge motor driver module	**1**	AliExpress Link	Amazon Link	*Upstream-listed driver. One channel drives the belt motor; the second channel is free for a second belt section.*

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CONTROLLER 1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Arduino Nano V3 ATmega328P CH340 with soldered headers	**1**	AliExpress Link	Amazon Link	*Upstream lists Uno, Nano or Mega. This build path uses the Nano on a breadboard. An Uno works with the same pins.*

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SENSORS 2 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	IR obstacle avoidance sensor module 3-pin adjustable	**1**	AliExpress Link	Amazon Link	*Part detector at the end of the belt (start/stop and counting). Adjust the potentiometer so it only sees parts on the belt.*
>>	KW12-3 micro limit switch with roller lever	**1**	AliExpress Link	Amazon Link	*Optional end-of-line switch for the pusher station. Skip if you do not build the pusher.*

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MOTION

1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	SG90 9g micro servo	**1**	AliExpress Link	Amazon Link	*Optional pusher/sorter arm, listed upstream as a compatible actuator.*

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POWER 2 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	5V 2A DC power supply 5.5x2.1mm barrel plug	**1**	AliExpress Link	Amazon Link	*Upstream lists an external 5 V supply. Powers the motor driver and servo; the Nano runs from USB while you test.*
>>	DC barrel jack 5.5x2.1mm female to screw terminal adapter	**1**	AliExpress Link	Amazon Link	*Brings the 5 V supply to the breadboard rails.*

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WIRING

2 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	400 tie-point solderless breadboard	**1**	AliExpress Link	Amazon Link	*Nano, driver and sensor wiring without soldering.*
>>	Dupont jumper wire set male-female male-male 40 pin	**1**	AliExpress Link	Amazon Link	*Sensor, servo and driver connections.*

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HARDWARE 3 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	M3 x 10 mm hex socket cap screws 50 pcs	**1**	AliExpress Link	Amazon Link	*Frame and bracket screws. Upstream does not list fasteners; M3 x 10 fits the frame holes in the model. Confirm count on your print.*
>>	M3 x 16 mm hex socket cap screws 50 pcs	**1**	AliExpress Link	Amazon Link	*Longer screws for the roller and motor bracket joints. Confirm lengths on your print; upstream lists no fasteners.*
>>	M3 stainless hex nuts 100 pcs	**1**	AliExpress Link	Amazon Link	*Nuts for the frame joints and sensor mounts.*

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CONSUMABLES 2 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	TPU filament 1.75 mm 95A 500 g	**1**	AliExpress Link	Amazon Link	*Belt links print best in flexible TPU. Consumable; printed geometry is in the package, not in this BOM.*
>>	PLA filament 1.75 mm 1 kg	**1**	AliExpress Link	Amazon Link	*Frame, rollers, guides and mounts. Consumable row; printed parts are not purchase rows.*

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WIRING 1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Mini USB cable 1 m data	**1**	AliExpress Link	Amazon Link	*Programming cable for the Nano (classic Nano uses Mini-USB; some clones use Micro or USB-C, check yours).*